

The Phospholipid Research Center: Advancing Phospholipid Science from Biomembranes to Pharmaceutical Research

Simon Drescher

Phospholipid Research Center, Im Neuenheimer Feld 515, 69120 Heidelberg, Germany

Phospholipids are essential components of biological membranes and play a central role in membrane structure, dynamics, organization, and function. Their physicochemical diversity, self-assembly behavior, and interactions with proteins, drugs, and other amphiphilic molecules make them highly relevant to both fundamental biomembrane science and pharmaceutical research.

This short introductory lecture will present the Phospholipid Research Center (PRC), an independent non-profit organization that supports scientific and technological research on phospholipids. Since 2006, the PRC has funded academic projects worldwide, particularly at the doctoral and postdoctoral level. While the PRC's main focus is on pharmaceutical and biomedical applications of phospholipids, including formulation science, lipid-based drug delivery systems, excipient functionality, and related technologies, selected projects also address more fundamental aspects of phospholipid behavior in biological and biomimetic membrane systems [1,2].

The presentation will briefly outline the PRC's mission, activities, and funding opportunities. By supporting excellent research and fostering exchange between academia and industry, the PRC aims to advance phospholipid science and to contribute to a deeper understanding of phospholipids in natural and engineered membrane systems.

References

- [1] S. Drescher, A. Blume, The 8th International Symposium on Phospholipids in Pharmaceutical Research – An update on current research in phospholipids presented at the biennial symposium of the Phospholipid Research Center Heidelberg, *Eur. J. Pharm. Sci.*, 210 (2025) 107126.
- [2] R. Wibel, P. van Hoogevest, S. Drescher, The role of phospholipids in drug delivery formulations – Recent advances presented at the Researcher's Day 2023 Conference of the Phospholipid Research Center Heidelberg, *Eur. J. Pharm. Biopharm.*, 197 (2024) 114215.